

A Recent Population History of Latin America and the Caribbean

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The claim

- Today, the total fertility rate (TFR) of Latin America and the Caribbean (LAC) is ≈ 1.5 .
 - Colombia's TFR in 2025: 1.01. Chile's: 0.99. Puerto Rico's in 2024: 0.87.
 - Only Haiti, Guyana, French Guiana and Saint Martin are above replacement, and Guyana and Saint Martin are probably only marginally above it and falling.
- Furthermore, the decline has been extraordinarily fast:
 - Jamaica went from 2.45 to 1.22 in 35 years.
 - Argentina went from 2.13 to 1.23 in 8 years.
 - Guatemala went from 3.8 to 1.9 in 20 years.
- The fall in the TFR is a world-wide phenomenon, but it has been particularly acute in LAC.
- Nobody forecast this. I certainly did not (so perhaps you should not listen to me!).

Outline

1. Where LAC was a few decades ago.
2. Where LAC is today, according to the countries' own vital registries.
3. Why the UN WPP overstates births and fertility.
4. Whether LAC is special.
5. What might account for it.
6. The economic consequences, above all slow growth and the political economy of aging.

Where LAC was

Two and a half centuries in one table

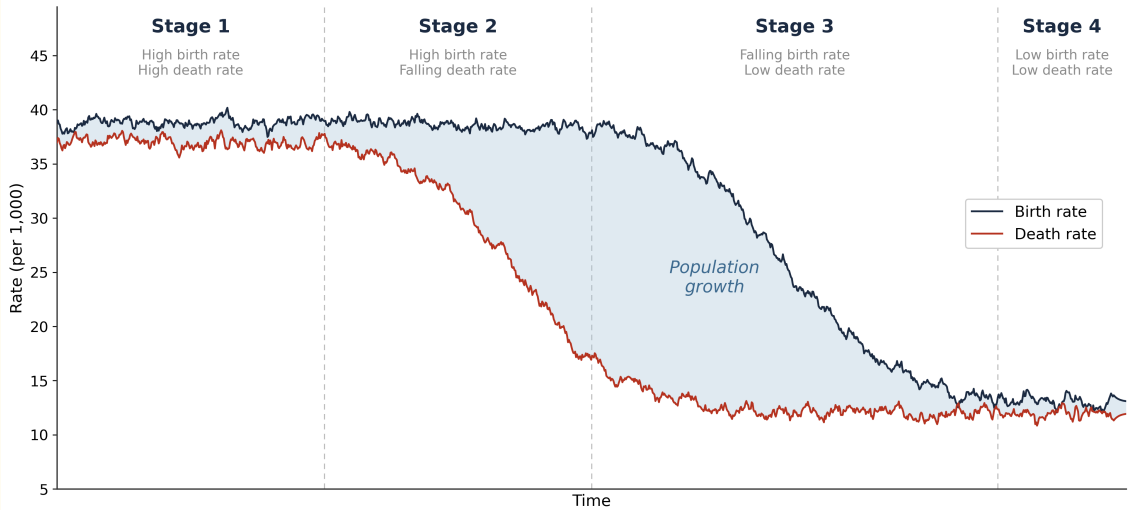
	1750	1800	1850	1900	1950	2000	2025
<i>Population (millions)</i>							
World	791	978	1,262	1,650	2,519	6,086	8,231
LAC	16	24	38	74	167	523	668
LAC, % of world	2.0	2.5	3.0	4.5	6.6	8.6	8.1
<i>Mean annual growth since the previous column (%)</i>							
World		0.1	0.5	0.5	0.8	1.8	1.2
LAC		0.3	0.9	1.3	1.6	2.3	1.0

- LAC grew faster than the world in **every** half-century from 1750 to 2000, and the gap kept widening: 2.3% a year against 1.8% between 1950 and 2000.
- The region went from **2%** of humanity to **8.6%** in 250 years. Then it stopped: since 2000 LAC has grown **more slowly than the world** and its share is falling, for the first time.

The postwar population boom

- In the early 1950s, the region's crude birth rate (CBR) was around **42 per thousand** and its TFR about **6** children per woman, barely below the start of the century and the level of the demographic ancien régime.
- What changed after WWII is mortality. Life expectancy went from **52 years** in the late 1950s to 71 by the late 1990s, and is above 75 today.
- The annual rate of natural increase peaked at **2.8% in the early 1960s**, the fastest any large region has ever sustained. At that rate a population **doubles in 25 years**.
- And people moved: **41% urban** in 1950, **86% today**. No region has ever urbanized so fast.
- This is why LAC, not Asia, was the textbook case of the population explosion: Ehrlich's *Population Bomb* in 1968, two decades of family planning programs pushed from Washington, and cities that grew faster than anyone could build for them.
- All the economic and political history of the region after 1945 is deeply mediated by this boom.

The “first” demographic transition



“The rule of 85”

- Take a country with life expectancy of about 85, the highest in the world today (Japan, Spain). Give it **1,000 births a year, every year**. What is its population in a hundred years? **$85,000 = 85 \times 1,000$** .
- So take any country's current births, multiply by 85, and you get *a sense* of its long-run population.
 - Costa Rica had **45,384** births in 2025. Long-run population: **3.86 million**. Current population: 5.2 million.
- Equivalently: **$1000/85 = 11.76$** . Once a country's CBR falls below 11.76 per thousand, births are already insufficient to hold population constant. This usually happens *after* the TFR falls below replacement.
- Also, notice that all of this is without migration.

A fast transition

- But since the boom was never about births rising, once births finally followed mortality, population growth collapsed.
- Furthermore, the transition, when it came, was **compressed**. What took Britain more than a century took Brazil and Mexico about 30 years.
- From a TFR near 6 in the 1960s to **below 2 by the 2010s**, and below 1.5 in half the region now.
- Births peaked in 1990 at **12.0 million** a year, which under the rule of 85 implied a region of **1.02 billion** people. Today they are around **8 million**, which implies **680 million**.
 - And LAC life expectancy is not 85 but about 76, which gives **608 million**, 8% *below* today's 668 million.
- ECLAC still projects the regional population to peak at **730 million in 2053**. I just do not see it.

Where LAC is

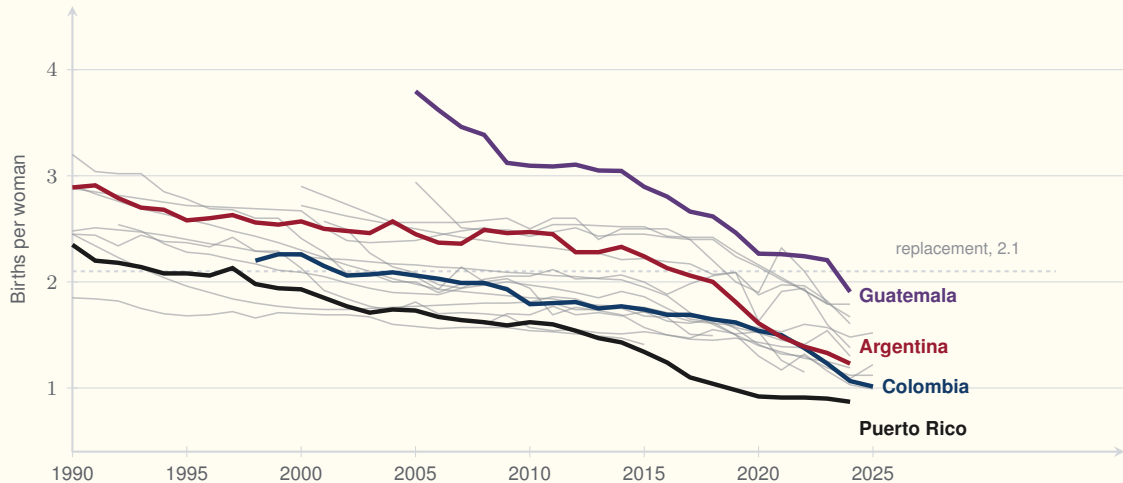
LAC today

- Latest year available from each country's **own registry**, not from the UN.

Country	TFR	Year
Puerto Rico	0.87	2024
Chile	0.99	2025
Colombia	1.01	2025
Costa Rica	1.12	2025
Bahamas	1.15	2022
Uruguay	1.19	2024
Jamaica	1.22	2025
Argentina	1.23	2024
Cuba	1.30	2024
Mexico	1.51	2025

Country	TFR	Year
Brazil	1.52	2025
Dominican Rep.	1.61	2024
Belize	1.63	2024
Peru	1.67	2024
Panama	1.79	2024
Nicaragua	1.80	2023
Guatemala	1.91	2024
Bolivia	2.06	2024
LAC	≈ 1.50	2024
U.S.	1.60	2024

LAC since 1990





How to read those panels

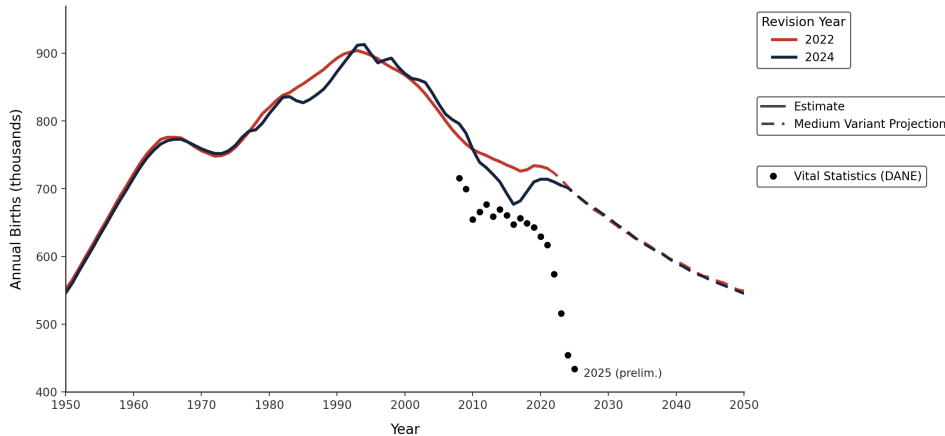
- Three things to notice:
 1. Almost every country crosses 2.1 between 2000 and 2020, and **none stops there**.
 2. The steepest declines are the most recent ones: Argentina, Chile, Colombia, Costa Rica and Uruguay all fall off a cliff after 2015.
 3. The countries that started highest are falling fastest. Guatemala lost 1.9 births per woman in twenty years.
- **Emigration compounds the problem**. Colombia and Spain share an ultra-low TFR of 1.0 to 1.2, but Colombians are moving to Spain. That cushions aging in Spain and accelerates it in Colombia.

Why not the UN WPP data?

Why I do not believe the WPP

- The WPP is the most cited global demographic dataset, and there is no good alternative. Most other projections are built on it. I use it too.
- But it is **systematically biased upward**, for three reasons.
 1. Its birth *estimates* exceed official vital registries, even where the UN itself rates registration as over 90% complete. Colombia in 2023: the registry recorded 515,549 births, the UNPD estimated 705,000, **37% too high**.
 2. Its *projections* rest on a mean-reverting assumption that produces an immediate rebound no matter how fast fertility is falling.
 3. Successive revisions keep moving **down**, and not fast enough.
- Both estimates and projections come from Bayesian hierarchical models with a nontrivial amount of analyst discretion.

Colombia: the UNPD and the registry



The 2024 revision moved the estimates down. It barely moved the projections.

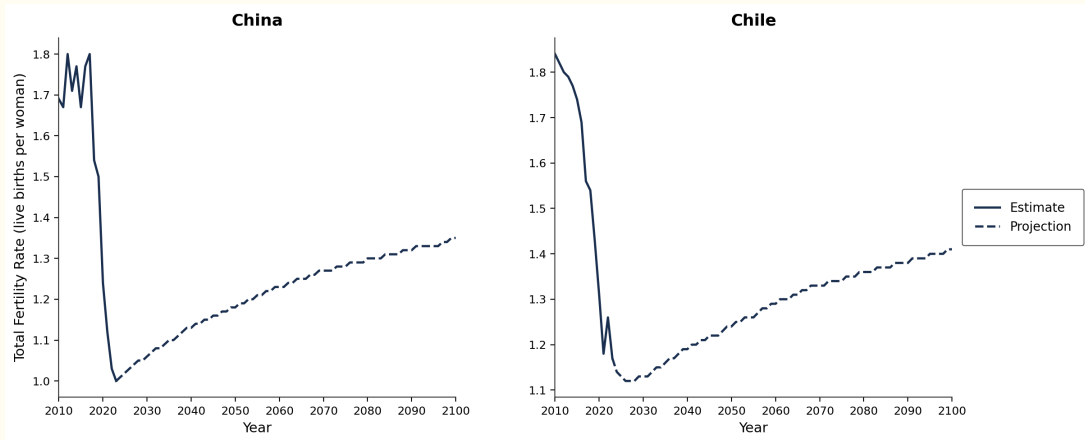
Where the rebound comes from

- Once a country is in the UNPD's "low fertility regime," its TFR follows a mean-reverting AR(1),

$$f_{c,t+1} \sim N(\mu_c + \rho_c(f_{c,t} - \mu_c), \sigma^2), \quad \mu_c \sim N(\bar{\mu}, \sigma_\mu^2), \quad \bar{\mu} \sim U[0, 2.1]$$

- That μ_c does all the work. South Korea rises monotonically from 0.72 in 2023 to **1.30** in 2100, China from 1.00 to **1.35**.
- And the turn is **immediate**. Between 2018 and 2023, China fell 0.54 and is projected to gain 0.05 by 2028. South Korea fell 0.23 and is projected to gain 0.07. The U.S. fell 0.10 and is projected to gain 0.01.
- Chile fell 0.37 over those five years and is projected to lose only 0.05 more.
- Demographers at the UNPD may think they are being clear about this. Very few economists know it.

China and Chile



The 2024 test: projections made in July 2024, for 2024

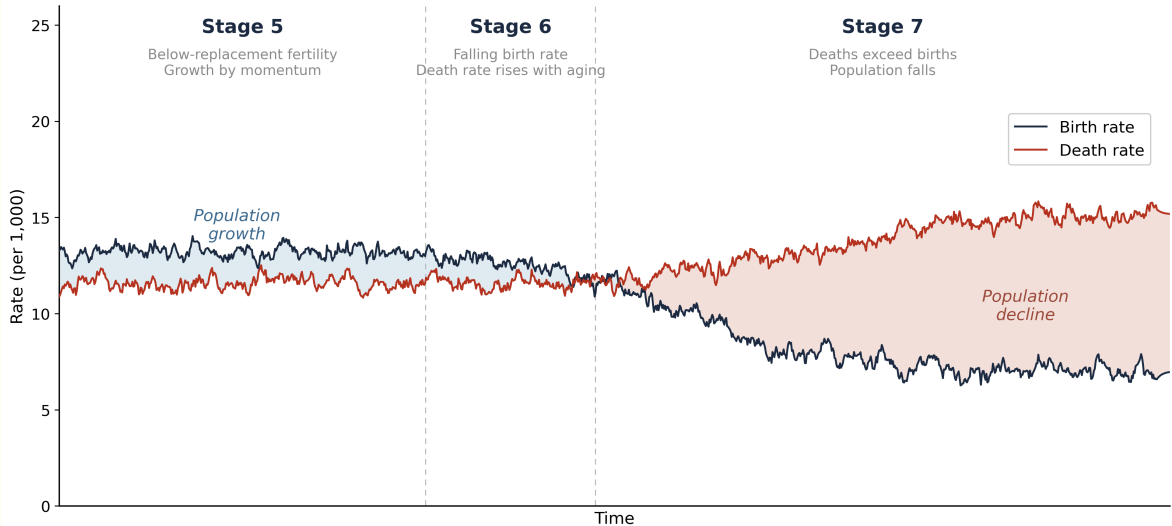
Country	Actual	WPP	Dev. (%)
Colombia	445,011	701,000	−36.5
South Africa	787,933	1,183,000	−33.4
Singapore	33,703	49,000	−31.2
Cyprus	9,766	14,000	−30.2
Rep. of Moldova	23,648	32,000	−26.1
Thailand	441,631	582,000	−24.1
Guatemala	299,256	380,000	−21.2
Poland	251,782	310,000	−18.8
Argentina	413,135	507,000	−18.5
Panama	59,729	72,000	−17.0
Albania	23,310	28,000	−16.8
Azerbaijan	102,310	121,000	−15.4
Türkiye	937,559	1,061,000	−11.6
Chile	154,441	172,000	−10.2
Uruguay	29,899	33,000	−9.4
Japan	686,173	751,000	−8.6
Hong Kong SAR	36,723	40,000	−8.2
Belarus	58,938	64,000	−7.9
Brazil	2,376,901	2,572,000	−7.6

Country	Actual	WPP	Dev. (%)
Slovenia	16,875	18,000	−6.2
Switzerland	78,256	83,000	−5.7
Spain	318,579	333,000	−4.3
Australia	292,318	305,000	−4.2
Greece	68,309	71,000	−3.8
Russian Federation	1,222,408	1,270,000	−3.7
United Kingdom	659,628	684,000	−3.6
Netherlands	166,143	172,000	−3.4
Denmark	57,079	59,000	−3.3
Republic of Korea	238,317	241,000	−1.1
France	629,000	636,000	−1.1
United Arab Emirates	106,915	108,000	−1.0
United States	3,628,934	3,654,000	−0.7
Uzbekistan	926,422	929,000	−0.3
Canada	365,737	361,000	+1.3
Kuwait	49,063	48,000	+2.2
Bosnia and Herz.	24,598	24,000	+2.5
Croatia	32,069	31,000	+3.4
Total	16,052,498	17,699,000	−9.3

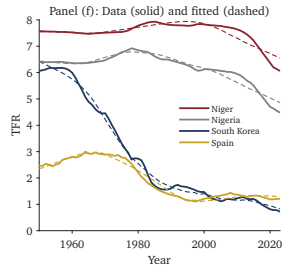
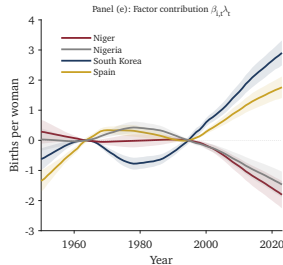
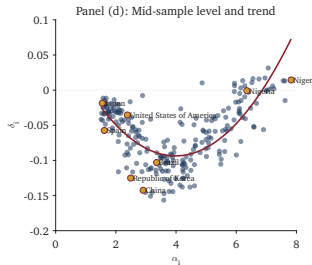
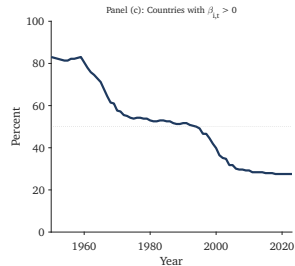
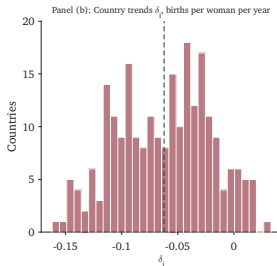
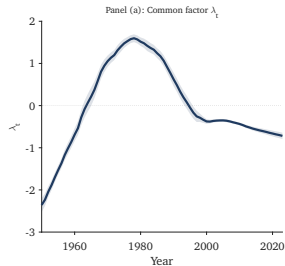
All countries with registries the UN deems at least 90% complete and population above one million. **Only 4 of 37** came in above projection. Absolute gap: **1.65 million births**.

Is LAC special?

The “second” demographic transition



Common factor and heterogeneous trends



The bar any explanation must clear

Two criteria

1. A **nearly universal** drop in fertility: almost every country carries a negative trend with no sign of leveling off.
2. **Country-specific timing**: the dates of the fast declines differ by decades across countries.

- Increasing cross-country dispersion is driven less by a single global force than by countries **moving through the transition at different times**.
- Most of the candidate mechanisms in the literature deliver one criterion or the other. Very few deliver both.

Mechanisms

The unpleasant integer arithmetic of replacement

- Before asking what causes the fall, notice how **low** the bar is. Children come in integers.

Children per woman	Cohort A	Cohort B
0	15	26
1	10	23
2	60	46
3	10	5
4	5	0
Completed fertility	1.80	1.30

- Cohort A is roughly many advanced economies until the early 1990s, and the U.S. until around 2008. Cohort B is **lowest-low fertility**.

Candidate 1: economic fundamentals

- **Quality-quantity** à la Becker-Barro. Many of the country trends are likely this, kicking in at each country's own pace.
- Delventhal et al. (2021): a Barro-Becker model matches the timing of transitions, including their acceleration, once augmented with the diffusion of skill-biased technology (Lucas, 2009), which does not arrive everywhere at once.
- This gets us part of the way. It does **not** get us the suddenness.
- It is hard to find movements in the skill premium large enough. No plausible calibration delivers Colombia's **35% fall in births in ten years** through this channel alone.

Candidate 2: smartphones

- Myers and Hooper (2026) exploit the fact that from June 2007 through February 2011 the iPhone was sold **only by AT&T**, identifying the effect from variation in that carrier's mobile broadband coverage.
 - They estimate the iPhone accounts for **33 to 52%** of the decline in the general fertility rate among women aged 15 to 44.
 - Hudson and Moscoso Boedo (2026) find the same for teen conceptions in England and Wales.
- My reading: smartphones **accelerated** the fall more than they caused it.
 - Italy's TFR was **1.19 in 1995**. In 2025 it was **1.14**. That gives you a sense of how much scope smartphones have as an explanation.
- Where they do matter is in the **diffusion of norms**. What took 25 years now takes 10.
 - Guatemala went from 3.8 children per woman in 2005 to **1.9 in 2025**.
 - Without smartphones, Guatemala would also have reached 1.9, just 20 years later.

A conjecture: modernity

- By **modernity** I mean a society organized around formal institutions, specialized expertise, and large-scale coordination.
 - Weber: modern societies organize economic life, law, and government through calculation, impersonal rules, and bureaucracy staffed by trained specialists.
 - Gellner, Giddens, Scott: complex societies become manageable only because knowledge is standardized and detached from local custom.
 - Mokyr: sustained growth became possible once useful knowledge was codified, accumulated, and widely shared.
- Modernity is **not** another name for capitalism. If anything the record points the other way.
 - Hungary fell below replacement in **1960**, Czechoslovakia in **1966**. Both had modest, short-lived baby booms in the mid-1970s; neither reversed the trend.
- I am aware this is more speculative than what economists usually entertain. I think it deserves the space.

Expensive third child, cheap childlessness

- A modern society makes the **third child expensive** and **childlessness cheap**, even at a fraction of frontier GDP per capita.
- Why children are expensive:
 - A child must be equipped to function in a world of formal knowledge and credentials: years of costly investment.
 - Children are no longer productive in the household, not even for small chores, having been replaced by “engines of liberation” now available to poor households too.
 - Modernity imposes tight career schedules, set by large organizations, that collide with the biological clock and with the kin networks that used to share the work.
- Why childlessness is cheap:
 - An impersonal society imposes next to no penalty on those who have no children. **Pensions and health care do the work that sons and daughters once did**, and lineage counts for little.

The gender mismatch

- Goldin (2026) pushes the line further: modernity changes gender roles **unevenly**.
 - Women's opportunities expand fast through education, work, and independence.
 - Expectations about housework and childcare change more slowly.
 - Particularly sharp as economies shift toward **services**, where women's comparative advantages are better rewarded.
- The mismatch operates on both margins: more women remain childless, and mothers have fewer children.
- This is attractive because it explains **East Asia and now LAC**: the societies where the mismatch is largest.
- It suggests a paradoxical reversal: societies with highly unbalanced gender norms **start** with very high TFRs, as men impose their preferences, and **end** with very low ones, as women resist.
- It also explains why cash transfers accomplish little while parental leave makes a difference. **The binding constraint is not the price of a child, but who does the work of raising them.**

Where modernity leaves each country

- Expensive children plus cheap childlessness gets us well below replacement, perhaps to 1.4. Recall the integer arithmetic: it does not take much.
- When each country arrives at 1.4 depends on how fast modernization runs, which is not the same thing as the level of GDP per capita.
 - That gives a nearly universal negative trend with growing heterogeneity: exactly the two criteria from the factor model.
- From 1.4, countries move in either direction:
 - Up to 1.6 or 1.7: higher income, better work-family balance, more gender equality, cheaper housing, less acute educational arms races, or a significant religious subpopulation. The U.S. has many, not all, of these.
 - Down to 0.7 or 0.8: lower income, no work-family balance, less gender equality, expensive housing, brutal educational competition, rapid secularization. China, Taiwan, and South Korea have all of these at once.
- In short: modernity in its current form seems largely incompatible with replacement-level fertility.

There is no feedback mechanism

- Economists trust markets because they self-regulate. Shortages raise prices, quantity demanded falls, new suppliers enter.
 - Not just theory: Smith (1962) showed double auctions converge with a handful of traders who know nothing about aggregate demand and supply. Gode and Sunder (1993) replaced the humans with machines bidding **at random** subject to a budget constraint, and allocative efficiency still reached almost 100%.
- **Nothing analogous pushes fertility back to replacement.**
 - No reason the positive consequences of a falling population (cheaper housing) should overcome the negative ones (slower growth, higher taxes) and restore a steady state.
 - And the welfare theorems break down in overlapping generations: generations far apart cannot trade, so the economy behaves as if markets were incomplete. The double infinity of agents and dated commodities kills the standard proof.
- Weil (2024) puts it best: **the replacement rate has no special status as an attractor or a target.** Nothing pushes a society to settle there.

Economic consequences

First, the good news

- **Teenage pregnancies.** Sharp declines in the U.S. over the last two decades, and around three-quarters of pregnancies among American teenagers are unintended. Much of this decline is welcome.
- **Resources and infrastructure.** Slower growth or falling population eases pressure on infrastructure, housing, and emissions, and makes sustainable resource policy easier to design.
 - LAC could redesign the cities that grew wildly, and often without decent services, during the boom of the 1950s to 1980s.
- **A window for emerging economies.** They can educate smaller cohorts better, rebuild infrastructure, and, before aging fully takes hold, enjoy some fiscal space for structural reform.
- **Climate.** Lower fertility might ease the transition to net zero. Integrated assessment models take population as an exogenous input, and their emission paths are sensitive to it in ways the literature has rarely explored.
 - Against this: the fiscal pressures of aging may make it harder to fund that transition.

The accounting of demographics

- Start from an identity:

$$\underbrace{y}_{\text{output}} = \underbrace{\frac{y}{l}}_{\text{labor productivity}} * \underbrace{l}_{\text{labor}}$$

- In growth rates:

$$\underbrace{g_y}_{\text{output growth}} = \underbrace{g_{\frac{y}{l}}}_{\text{labor productivity growth}} + \underbrace{g_l}_{\text{labor growth}}$$

- These definitions discipline what we can expect. Everything that follows is arithmetic, not a model.

The way we were

- The U.S. around 2000, in round numbers close to the actual ones:

$$\underbrace{3.0\%}_{\text{output growth}} = \underbrace{2\%}_{\text{labor productivity growth}} + \underbrace{1\%}_{\text{labor growth}}$$

- This is why most economists around 2000 took 3% as the standard for measuring economic performance:
 1. Booming, unemployment falling and labor growing faster than average: you see 4%.
 2. Depressed, unemployment rising and labor growing slower than average: you see 2%.

The way we will be

- The U.S. around 2050, holding labor productivity growth fixed and assuming no net immigration:

$$\underbrace{1.5\%}_{\text{output growth}} = \underbrace{2\%}_{\text{labor productivity growth}} + \underbrace{-0.5\%}_{\text{labor growth}}$$

- 1.5% is the new normal. A boom delivers 2.5%, a slump 0.5%.
- The Fed can do nothing about this with monetary stimulus, and neither can the federal government with fiscal packages.
- A shrinking labor force is **not a shortfall in aggregate demand**. It is a drop in the long-run trend of potential output.
- And, as we will see, the 2% assumption is itself generous.

Japan is doing about as well as it can

1991-2023	Canada	France	Germany	Italy	Japan	Spain	UK	U.S.
GDP	2.40	1.56	1.27	0.79	0.79	1.92	2.13	2.60
GDP per capita	1.26	1.07	1.14	0.67	0.78	1.24	1.56	1.69
GDP per work.-age adult	1.38	1.28	1.41	0.92	1.33	1.28	1.63	1.73
GDP per hour worked	1.09	1.07	1.37	0.67	1.22	0.58	1.45	1.62
GDP per worker	0.91	0.82	0.90	0.48	0.62	0.39	1.34	1.55
Population	1.12	0.49	0.13	0.12	0.01	0.68	0.55	0.90
Working-age adults	1.00	0.28	−0.14	−0.13	−0.53	0.63	0.48	0.85
Total hours worked	1.34	0.50	−0.10	0.14	−0.43	1.35	0.66	0.98

- Average annual growth rates in percent. Working-age adults are 15 to 64.

Swap the demographics

- Using data per hour worked:

$$\underbrace{0.79\%}_{\text{Japan's output growth}} = \underbrace{1.22\%}_{\text{Japan's productivity growth}} + \underbrace{-0.43\%}_{\text{Japan's labor growth}}$$

$$\underbrace{2.60\%}_{\text{U.S. output growth}} = \underbrace{1.62\%}_{\text{U.S. productivity growth}} + \underbrace{0.98\%}_{\text{U.S. labor growth}}$$

- Now hold productivity fixed and switch labor growth between the two:

$$\underbrace{2.20\%}_{\text{Japan, counterfactual}} = \underbrace{1.22\%}_{\text{Japan's productivity growth}} + \underbrace{0.98\%}_{\text{U.S. labor growth}}$$

$$\underbrace{1.19\%}_{\text{U.S., counterfactual}} = \underbrace{1.62\%}_{\text{U.S. productivity growth}} + \underbrace{-0.43\%}_{\text{Japan's labor growth}}$$

- The verdict **reverses**. Instead of the U.S. leading Japan by 1.8% a year, the U.S. lags by 1.0%.

Will productivity growth hold at 2%? Ideas

- Semi-endogenous growth: the long-run growth rate of income per capita is proportional to the growth rate of the **number of researchers** (Jones, 1995). Researchers cannot grow faster than population in the long run.
- The logic runs deeper than accounting. Ideas are **nonrival**: smaller populations produce fewer ideas *and* spread their cost among fewer users.
 - Jones (2022): an economy with permanently negative population growth converges to a path where knowledge and living standards **stop growing altogether**. The *empty planet*.
- And ideas are getting harder to find. Bloom et al. (2020): U.S. research productivity falls by roughly **5% a year**. We need an ever larger number of researchers just to hold growth constant.
- A country with a TFR of 1.2 will not have many pathbreaking scientists and engineers.

Not just the number of researchers: the top of the distribution

- South Korea: 1,080,535 births in 1960 against 230,028 in 2023. A 4.7-fold decline in cohort size.
- What does that do to the *most talented individual* in each generation?
- **Gaussian tail**: not much. The top child born in 1960 is expected to be 4.84σ above the mean; the top child born in 2023, 4.53σ .
 - Nearly a million additional births buy less than a third of a standard deviation at the very top.
- **Pareto tail with $\alpha = 2$** : very different. The frontier scales with the **square root** of cohort size. The median of the cohort maxima is 883 times the median person in 1960, against 407 in 2023.
- Which case applies? The evidence on returns to innovation points to **fat tails**, and specific individuals matter beyond their number (Azoulay et al., 2010).
 - If growth depends on a handful of exceptional innovators, shrinking cohorts hurt most.

But don't we care about output per capita?

- Yes and no.
- **Yes:** output per capita is the right measure of individual welfare. Slower labor force growth costs us less than the headline numbers suggest. Japan has kept improving its living standards, and simulations for the U.S. suggest the same (Weil, 2026).
- **No:** our capacity to service public debt and meet social security obligations depends on **total output**. Bondholders and retirees are paid in levels, not in ratios.
- And in a world of great-power competition, **defense spending requires total output**, not output per capita.
- Second-round effects I cannot explore here: inflationary pressure as governments try to monetize debt, real interest rates, housing prices, and the internal redistribution of population, with a few large “sponge cities” absorbing a dwindling population.

The political economy of aging

- Many economists forget that the **first-order** problem of an aging society is how to redistribute income from workers to retirees. How fast output grows is, surprisingly, the lesser question.
- With roughly as many retirees as workers, and approximate parity of income, **what each worker produces must be split in two**.
 - It does not matter whether we do it through taxes, as in a pay-as-you-go system, or through capital income, as in a fully funded one.
 - Workers will not be happy to see so much of GDP channeled to retirees.
- This is likely to poison the political system and lead to dysfunctional policies. We already see it in Western Europe's fights over retirement benefits, which make structural reform harder.
 - Much of the American unhappiness about the profits of large corporations, largely owned by the old, is the **same conflict** in the fetishized form of equity ownership.
- "AI will fix this" misses the point. The redistribution problem is about **politics**, not productivity growth.

And Japan was the easy case

- The Japanese slowdown we documented is arguably the **best-case scenario**.
 - A homogeneous society with high levels of consensus.
 - Strong institutions.
 - High income per capita.
- Now imagine **Mexico or Tunisia** facing the same pressures in 30 or 40 years, with low-capability states, less consensus, a tradition of political conflict, and a fraction of the income per capita.
- Middle-income countries will not have the margins of maneuver Japan has.

Concluding remarks

- When I open my undergraduate course on world economic history, I argue that history teaches us lessons about today's economy.
- That is **not true of population**. We have never seen TFRs as low as today's.
- Will they recover? I do not know. My best guess is a small rebound from tempo effects and, short of a large change in society, not much more.
- **LAC**: the gender mismatch plus weak growth prospects. I expect something around 1.2.
- And LAC gets the problem twice: falling births *and* the young adults who leave.

Where that leaves LAC

- A region that will age **before it gets rich**.
- Not knowing the causes, I cannot forecast fertility over the coming decades. And forecasting is the **easy** part compared with designing policy.
- Economists are used to markets that correct themselves. **Nothing in an economy pushes fertility back to replacement.**
- All of this could change with AI or with shifts in population composition. We will see.
- But on current evidence, deep sub-replacement fertility is the “**new new normal**,” and unless we find a way to reorganize our societies, we had better learn to handle it.
- We do not have a map. What we have is birth numbers, and those are sobering. **In these waters, the pretense of knowledge is the most dangerous temptation.**